Appendix G: Pilot Car Load Movement Guidelines

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Pilot Car Load Movement Guidelines

2020

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Acknowledgements

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Section 1: Introduction

1.1 Purpose of the Guidelines

Pilot cars play an important role in making sure that unusually large loads or wide vehicles travel BC's roads safely. Pilot cars function to warn other road users of the presence of an oversize vehicle, assist in maneuvering the load through constrained areas, and keeping traffic delays to a minimum.

These guidelines have been developed to support activities for the safe movement of oversize loads over BC highways. The standards for pilot cars and their equipment are set out in the *BC Commercial Transport Regulations Division 8*. This manual is intended to clarify, enhance and support the conditions for travel that are set out in provincial permits for oversize and overweight loads.

The traffic control described and illustrated in these guidelines is generally the minimum required. No one standard sequence of signs or other control devices can be set up as an inflexible arrangement for all conditions and locations, due to the variety of conditions encountered. It should also be recognized that while the Pilot Car Load Movement Guidelines contain mandatory language such as "shall" there may be circumstances where strict compliance with such requirements is not reasonable and it will be necessary to deviate from the requirements.

Loads that are very large or very heavy often require the development of detailed, written Transportation Management Plans in the course of planning and seeking approval for the necessary permit to move the load. In those cases, in addition to the content of these guidelines and the legal requirements from the BC Commercial Transport Regulations, please consult the guidance in *Section 6.4.4 Chapter 6, of the Commercial Transport Procedures Manual.*

1.2 Definitions

Lead pilot car: A pilot car travelling ahead of the load. Where there are two pilot cars ahead of the load, the lead pilot car is the one closest to the load. The lead pilot vehicle should travel 4 to 8 seconds ahead of the load. The purpose of the lead pilot car is to warn oncoming drivers that the load is approaching and to double check clearances and other constraints for the load.

MoT: BC Ministry of Transportation and Infrastructure

Multilane highway: A highway with at least 4 lanes (typically 2 lanes in each direction).

Oversize load: The vehicles and loads that are overheight, overwidth, overlength and/or overweight, i.e. any load or vehicle for which the terms of a permit or authorization requires the use of one or more pilot cars.

Pilot car: A pilot car, for the purposes of these guidelines, is a vehicle that is escorting commercial transport vehicles. A pilot car used to escort commercial transport vehicles is different than a pilot car used to lead traffic through a work zone.

Rear pilot car:	The rear pilot should travel 4 to 8 seconds behind the load. The purpose of the rear pilot car is to warn drivers approaching from the rear, to monitor the load (e.g. cargo securement, off tracking, etc.), and to notify the load driver if any vehicles may be passing. The rear Pilot Car should avoid getting in the load driver's blind spot without communicating with the driver.
Scout pilot car:	When there are two pilot cars travelling ahead of the load, the scout pilot car travels further ahead in order to identify potential issues before the front pilot car and load arrive. The scout pilot car is also responsible for establishing appropriate traffic control, if required.
Traffic control:	The act of slowing, stopping, or directing general purpose traffic on a highway (flagging). All traffic control must be conducted by a certified traffic control person as per the Occupational Health and Safety Regulation, Division 18. When a Pilot Car is only operating as a mobile warning device for the load, it is not conducting traffic control.
Trailing pilot car	When there are two pilot cars travelling behind the load, the pilot car furthest away from the load is the trailing pilot car. The trailing pilot car position is typically a temporary position that a second or third pilot car may occupy following a traffic control situation where traffic was held in order to let the load manoeuver.
Transportation Management Plan	A transportation management plan is used to plan and implement moves for loads that are very large or very heavy. They detail the specific agreed processes for the movement of the load. This is different from a “traffic management plan” which is used to outline vehicle movement through a construction zone.
Two and three lane highway:	A highway with one lane in each direction but may have sections with passing or truck climbing lanes resulting in a total of three lanes.

1.3 The Importance of Pilot Cars

Provincial highways are designed to fit standard vehicles with standard dimensions. However, there are times when the movement of oversize loads are required to meet the needs of both public and industry.

Oversize load vehicles and the loads they carry are often too long, wide, or high for the marked lanes of a highway or the infrastructure on and over the highway. This can create operational problems, especially as many of British Columbia’s highways are two-lane, two-way highways.

Pilot cars ensure that all road users are aware of the potential hazard, and know what to do if they encounter these vehicles so they can pass safely.

Section 2: General Pilot Car Operations

2.1 When Pilot Cars are Required

Pilot Car requirements for oversize load moves on provincial highways will be established by the terms of the permit issued for the move. These permits also establish other terms for the move, including time of day movement restrictions.

One or two pilot cars may be required when the load width, length, or height result in needing additional roadway space to maneuver. Basic rules for whether one pilot car should be in front of or behind the load are set out in the Commercial Transport Regulations, section 8.08. When two pilot cars are used, generally one is positioned as lead and the other follows the load as the rear pilot car. The main duties of the rear pilot car driver are to communicate with the load driver about surrounding traffic or other obstacles, and to monitor the load.

Three or more pilot cars may be required for more complex moves. The requirements and typical positioning of the pilot cars would be set out in the permit conditions and may vary depending on the locations and situations along the route. A third pilot car would typically assume the scout pilot car position and may travel well ahead of the load and lead pilot car in order to identify appropriate traffic control locations or pinch points ahead of time.

CVSE 1000 and other T-forms set out the general pilot car requirements. The load permit may have additional requirements. If a transportation management plan has been required as a condition of approval for the move, it may have further pilot car and traffic control requirements.

Figure 2.1 – Sample of permit pilot car requirements

Pilot Cars:
Pilot cars must have two-way communication with the driver of the oversize load.

	Width →	Up to 3.2 m		3.21 to 3.5 m		3.51 to 3.8 m		3.81 to 4.4 m	
		Day	Dark	Day	Dark	Day	Dark	Day	Dark
Length	Up to 27.5 m	0	0	1*	1	1	1	1***	2
	27.5 to 31 m	1	1	1	1	1	1	2	2
	31.1 to 36 m	1	2	1	2	1	2	2	2
	36 to 40 m	2**	2	2**	2	2	2	2	2

* Or the towing vehicle must be equipped with and operate one or two amber flashing lights
** Up to 3.5 m wide, only one pilot car is required for travel over four (4) lane highways
*** Two pilot cars for travel on Highway 97 from the junction of 77/97 to the Yukon Border
For weight permits, where the approval to obtain a permit includes bridge crossing conditions, a minimum of two pilot cars are required at each bridge crossing condition location

Additional Pilot Car Requirements for Front or Rear Projection:

- If you are using an allowance for front projection beyond 3 m of the kingpin or beyond 6.5 m measured forward of the turn center of the front steering axle group, a minimum of one pilot car is required (unless otherwise provided for by commodity or vehicle policy).
- If you are using an allowance for rear projection beyond 6.5 m from the turn center if the rear-most axle group, a minimum of one pilot car is required (unless otherwise provided for by commodity or vehicle policy).
- A minimum of two pilot cars will be required if the load exceeds 3.8 m in width and the front projection is in excess of 3 m and/or the rear projection is in excess of 6.5 m.

For overall widths over 3.2 m, a minimum of one pilot car is required:
On Vancouver Island:
- Highway #4 from the Taylor River Bridge (40 km west of Port Alberni) to Tofino
- Highway 30 from Highway 19 to Port Alice

On the Sunshine Coast: Highway 101 from Langdale to Earl's Cove
In the Lower Mainland/Southern Interior: Highway 99 from Pemberton to Lillooet
In the Kootenays:
- Hwy 3A and 31 (Balfour-Kaslo-Galena Bay Highway)
- Hwy 33, southbound only, to approach and cross the Westbridge Bridge

In the Peace River Region: Highway 97 from Junction 77/97 to the Yukon border

2.2 When Certification is Required

In BC, pilot car operators are not required to be certified to do the parts of their work that do not involve controlling traffic, such as acting as a moving visual warning device escorting a commercial load, and communicating with the load driver to assist in the safe movement of the load.

However, certification is required to perform traffic control in a high risk situation, such as on a highway. In BC, the basic level of certified traffic control training is available through the *BC Construction Safety Alliance,* and the Occupational Health and Safety Regulation further requires that employers must provide additional training if the traffic control needed is more advanced than what is covered in the basic course. The basic course teaches the current standards for traffic control, primarily intended for flagging at road construction sites, and also covering information about equipment, flagging movements, stopping distances, etc. These guidelines establish the movement of commercial loads using pilot car escorts and standards for advanced traffic control training related to the movement of commercial loads.

Further, oversize and overweight permits in BC require that traffic control, where necessary, must be done by a certified traffic control person using the methods set out in these guidelines, or by a peace officer.

Out-of-province pilot car operators who engage in traffic control in British Columbia shall either be certified traffic control persons in British Columbia or proof of valid traffic control person certification from their home jurisdiction.

2.3 Related Laws and Policies

Pilot car use is one piece of the overall permitting process for oversize and overweight loads. Some other tools are:

- Standards for weights and dimensions of vehicles and loads without permits on provincial highways are in the BC Commercial Transport Regulations, *Division 7*.
- Legal requirements for pilot cars and their equipment are in the BC Commercial Transport Regulations, *Division 8*.
- The *Commercial Transport Procedures Manual* which sets out policy guidelines for permitting
- *T-Forms* (permit attachment forms) which are attached to permits to provide travel times, numbers of pilot cars required, basic light and flag information, and sometimes routing assistance
- *Extraordinary Load Approvals* and *CVSE1052* forms for extremely large or heavy loads
- In BC, permits are issued through the Provincial Permit Centre, 1-800-559-9688, and online at *OnRouteBC*. Permit Centre hours are 6 am to 10 pm, 7 days per week, except Christmas Day.

2.4 Best Practices

Pilot car operators need sufficient driving experience and training that they can manage the complexities of assisting the load they are escorting in addition to the usual demands of operating a motor vehicle. Keep in mind that pilot car operators must:

- Comply in all respects with provincial or other government laws for the movement of vehicles.
- Drive defensively – anticipate potential incidents and plan for them. Expand the area you would be aware of when operating your personal vehicle to the area the load driver needs to be aware of.
- Be aware of the load driver's lines of sight.
- Keep windows uncluttered, and mirrors properly adjusted.
- Respect the weather and the road conditions, and drive accordingly. Remember that, in some cases, permits become invalid when road conditions are poor.

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Section 3: Load Movement Planning

3.1 Non-provincial Highways

The use of non-provincial highways such as municipal streets and industrial roads is subject to approval and/or permitting by the jurisdiction operating those roadways, and operational requirements may differ. Sometimes those jurisdictions place controls (such as noise bylaws) that affect oversize load moves, and must be planned for in advance.

3.2 Route Planning or Route Survey Guidelines

Single trip oversize or overweight permits for loads that require the use of pilot cars include approved routing information. The approved route must be followed exactly, unless a peace officer directs otherwise.

When using a term permit (which does not have a specified route), or if assisting a carrier to plan a route before a single trip or term permit is obtained, consider the following:

- What are the load dimensions? Are there areas along the route which will require special attention?
- Known constraints on the highway, and how to manage the traffic. For example, if oncoming vehicles will have to be stopped to allow the oversize load to pass, plan a suitable place for directing road users at constraint points along the route. Very large loads may need approval of a Transportation Management Plan by MoT, RCMP and utility companies before a permit can be issued, and these moves often involve making arrangements ahead of time to temporarily move infrastructure such as signs and barriers. Guidelines for developing Transportation Management Plans can be found in *Section 6.4.4 Chapter 6* of the Commercial Transport Procedures Manual.
- Where are the pullouts and other stopping points along the route?

The MoT maintains an overheight registry of major structures located along the numbered highways in BC, available at *www.DriveBC.ca*. Carriers and/or pilot car operators should use this tool during pre-trip planning to determine the route to request for the permit. Clearances should be checked for each and every trip.

3.2.1 Using Height Poles on Route Surveys

From time to time, a pilot car with a height pole may be engaged to double-check route viability during the planning stage for an oversized move. Before embarking on such a trip, height clearances on major routes should be checked using the CV Height Clearance Tool at *www.DriveBC.ca*. If the pilot car operator engaged to do route analysis does not have strong local knowledge, they must check with local weigh scale staff before running the route, to identify known issues that might prevent the trip.

Power and other overhead lines and bridge structures may have lower road clearance tolerances on secondary highways and municipal roads, and care must be taken at all times, including during route analysis, to maintain a safe clearance distance from the lines. See *Section 4.2.1* for height pole requirements.

If a visual estimate indicates that the height pole may come in contact with a line, stop in a safe location and assess the situation. Use a safe tool to measure overhead wires and lines or traffic signals - preferably a laser or other electronic measuring device. If using a measuring stick, it must be of a non-conductive material, and must not make direct contact or come too close to the line or signal; measure to the side. Any measurement activities that might interfere with the flow of traffic must be done with appropriate approvals, using certified traffic control personnel.

www.DriveBC.ca is a good resource for planned construction activity on the selected route. It can also be helpful to use web tools such as maps that have street view to assist in determining the shape of a structure you will be encountering.

Note: Legal height in BC is 4.15 m, measured from the surface of the roadway. In order to exceed 4.15 m with a height pole, except while escorting a permitted load, an oversize permit is required (Provincial Permit Centre – *onRouteBC.gov.bc.ca* 1-800-559-9688)

In order to exceed 4.88 m overall height (5.33 m in the Peace River Area), an *Extraordinary Load Approval* and a signed *CVSE1052* form are required before the permit can be obtained. Please allow a few days to obtain these documents.

3.3 Expect the Unexpected (Contingency Plan)

Develop and review contingency plans with the pilot cars involved with the move, the load carrier and their oversize load driver

- Plan for vehicle breakdowns
- Plan for emergencies, including allowing emergency vehicle to pass
- Plan for sudden weather changes
- Plan for contact with an overhead obstruction
- Plan for railroad crossing issues – See *Section 8.5* and *Operation Lifesaver Tips for Professional Drivers*
- Plan for vehicle collisions/accidents and property damage
- Plan for allowing scheduled vehicles to pass (Bus Lines, Mail Services, etc.)
- Cell coverage maps from cell providers should be checked to determine where cell coverage areas may be lost
- Plan for bringing in additional or replacement Pilot Cars
- If the pilot car operators are not certified for traffic control, plan for how certified traffic control persons will be brought in if an unexpected circumstance will require traffic control.

3.4 Pre-trip Meetings

A safe move requires good teamwork. Before beginning an oversize load move, make sure all contacts required for the move are clearly identified with contact information. Discuss and plan the move with the load driver and any other pilot car operators or teams involved.

Prior to the start of an oversize load move, a team coordination meeting should take place. Meetings should be held on the initial move day, and on subsequent days if the move takes place over several days.

- Discuss roles and responsibilities of the move's team members
- Discuss safety precautions and communications to be used during the move and ensure all team members have two-way radio and cell phone communication for the move (see *Section 4.4 – 4.7*)
- Verify the oversize load dimensions against the permit before leaving
- Review the route plan and verify the route hazards expected to be encountered that day (bridge and overpass clearances, signs, wires, concrete barrier and shoulder issues)
- Check the permit and any attached approval for bridge crossing or other locations that have mandatory traffic control, and ensure that those locations are known and understood
- Discuss and complete a Job Hazard Safety Analysis

A Job Hazard Safety Analysis is part of the team coordination meeting and consists of the following steps:

- Identify load-specific risks
  - Fire
  - Explosive potential
  - Dangerous goods (identify railroad or tunnel restrictions based on the material to be moved)
  - Load configuration (protruding components, weight/load shift potential)
  - Fragile or collapsible loads
  - Time sensitive or perishable materials
- Ensure appropriate emergency equipment is on hand and team members know how to use it
- Review emergency procedures
  - Review procedures for communications during emergencies
  - Review procedures to delay or abort the move
- Review load dimensions and the subsequent limitations
  - Ground clearances
  - Load Height
  - Maneuverability limitations
- Review contingency plans for emergencies
  - Contact information to emergency providers along the route
  - Stopping sites for reviewing load security and breakdowns

3.5 Checks just prior to Moving the Oversize Load

- Check the pilot cars to ensure they are equipped with the proper signage, communication, and safety equipment for the move.
- Test communication equipment prior to the load proceeding.
- Verify the dimensions of the load and vehicles involved in the move to ensure load and vehicle fit within the clearances of the expected structures and roadside geography expected along the route.
- Identify any features of load security that pilot car operators might help monitor, if needed.
- Verify all permits are available to be shown to officials where required.
- Verify the permit restrictions, including the times the move may take place.
- Verify the route outlined on the permit matches the planned route.
- Review how each vehicle in the move will be positioned for the move.
- Identify the location where the next safe stop will be made, if required.

Section 4: Equipment and Communication

4.1 The Pilot Car

Vehicle requirements and the specifics of signs and lights required on the pilot car may be found in *Division 8* of the Commercial Transport Regulations.

We recommend choosing a vehicle, such as a light pickup or SUV, that:

- has good stability at highway speed with a sign displayed;
- allows the sign to be clearly visible in heavy traffic;
- offers sufficient cargo capacity for required equipment, signs and devices.

4.2 Pilot Car and Driver Equipment to be Worn/Carried

As a minimum, you must wear and carry/use the following:

- Safety apparel as per *BC MoT Technical Circular T-09/05*
- Required signs and devices, as described in section 4.3
- Radios and cell phones, as described in section 4.5

Recommended additional equipment:

- First aid kit
- Fire extinguisher
- Tape measure or other measuring equipment (minimum 8 m/25')
- Personal items: Drinking water, food, extra clothing, blanket, medication
- General tool kit with pliers, wrenches, screwdrivers, etc.
- Jumper cables
- Motor oil, coolant, windshield fluid, etc.
- Extra "Oversize Load" and "Wide Load" or "D" sign, meeting the requirements from *Section 8.03 of the Commercial Transport Regulations*, with hardware for attaching it to a load

4.2.1 Height Pole

Height poles, if used, must be made of non-conductive material such as fiberglass or plastic, and equipped with a flexible tip. The height pole should be set at 100 to 150 mm above the height of the load to accommodate flexing in the pole caused by wind resistance when travelling. The intent is not to strike power lines or overpasses, as height on the route should always be known in advance of the move. The height pole is intended to be an extra safety measure in case of unexpected clearance issues on the route.

Scout and/or lead pilot cars may be equipped with height poles, under oversize permits, and preference would be given to the scout car to accommodate stopping distance for the load. If using a height pole on a lead pilot car, and a potential height obstacle is observed ahead, the lead pilot car and the load should develop a large enough gap to allow the load to stop if necessary.

If height pole(s) are used on scout and/or lead pilot cars while escorting overheight loads travelling under a single trip permit, no additional oversize permit is required for the pilot car(s) during the permitted trip.

4.3 Signs and Devices for Traffic Control

When traffic control is needed for 15 minutes or more, additional devices will be needed for traffic control outside a vehicle. For example, sometimes unforeseen circumstances, such as a breakdown, can result in lengthy unplanned traffic control. Information on the following signs and devices can be found in *Chapter 4 and Appendix B of the TMM.*

With each oversize load move requiring pilot cars, at least the following signs and devices must be carried:

- One (1) Stop/slow C-027 paddle per crew member - (Illuminated traffic stop paddles, with LEDs around the perimeter of the sign, are an acceptable enhancement for stop/slow C-027 paddle as per *WorkSafeBC Occupational Health and Safety Regulation Guidelines Part 18, Section 9a*)
- One (1) illuminated baton with light that appears red or flashlight fitted with a red signalling wand, per crew member
- Two (2) Road Work Ahead C-018-3A signs, in fluorescent pink or fluorescent orange, with necessary supports
- At least five (5) channelizing devices (tubular markers, barrels, or cones), to form tapers

This equipment may be distributed among all the pilot cars involved in the move.

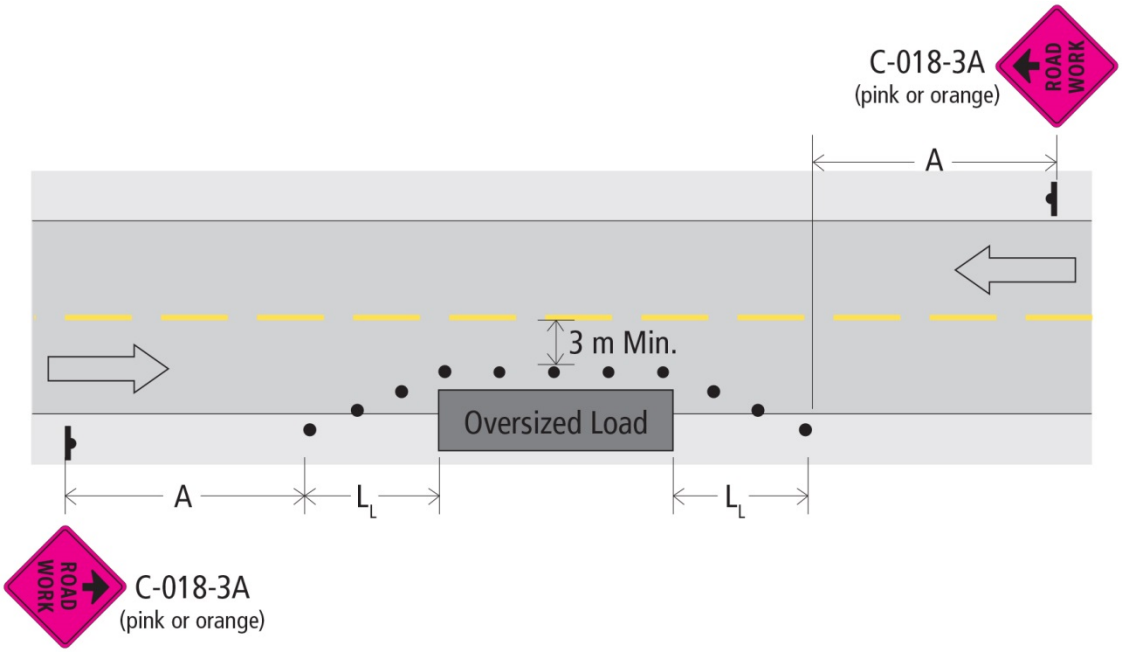
> Note: The above signs and devices are mandatory for all oversize/overweight load moves requiring pilot cars, even if the pilot car operators are not certified traffic control people, in case of an emergency situation.

Pilot car crews should also carry the following additional equipment, especially crews escorting very large or very heavy loads.

- Two (2) Traffic Control Person Ahead C-001-1 signs
- Two (2) Prepare to Stop C-029 signs
- Temporary sign support for each sign
- Five (5) additional channelizing devices to form tapers

Pilot car operators should be prepared to set up simple layouts such as that shown in *Figure 4.1* for situations such as vehicle breakdowns or unexpected stops. If the anticipated incident duration is greater than 15 minutes, or if a complete lane closure is required, certified traffic control people must be brought on scene and additional signage and devices may be required.

Figure 4.1 – Emergency or Incident – Load Pulled Over but Partially Encroaching into Travelled Lane



Refer to Table A and Table B in *Section 6.6 of the Traffic Management Manual for Work on Roadways* for spacing A and L_L.

4.3.1 Vehicle-mounted Stop Signs (Optional)

A pilot car may be equipped with a vehicle-mounted stop sign, mounted approximately midway down the vehicle, on the driver's side, at such height that it is visible to approaching vehicles and does not block the pilot car driver's mirrors or their view from the mirrors. The vehicle-mounted stop sign must swing out and in under the control of the driver, and must be covered at all times except when escorting an oversize load. The appropriate sign for this use is a double-sided, 600 x 600 mm, R-001 stop sign.

Illumination, such as LED lights around the perimeter of the sign, is acceptable. Red or amber alternately flashing lamps, such as those used on school buses, are prohibited on vehicle-mounted stop signs for pilot cars.

See *Section 5* for guidance about when a vehicle-mounted stop sign may be used in traffic control.

4.3.2 Dynamic Message Signs (Optional)

In addition to the pilot car sign mandated in Division 8 of the Commercial Transport Regulations (e.g. Wide Load, Oversize Load, D), dynamic message signs (DMS) may be used to direct traffic and provide additional messaging. DMS may display traffic control message such as STOP, SLOW, DO NOT PASS or OBEY FLAGGER.

When used at night, a DMS shall adjust brightness levels to maintain legibility and visibility for oncoming vehicles.

DMS should use primarily yellow text. Other text colours may be permissible, e.g. STOP messaging in red text, but its use should be limited. For nighttime visibility, coloured text on a black background is preferred.

Figure 4.2 Dynamic Message Sign



4.4 Communication during the Move

One of the pilot car operator's most important roles is to monitor the area around the load and communicate what they see to the oversize load driver. Pilot cars in front of the load are especially helpful for communicating about obstacles ahead, and those to the rear are very helpful for keeping the load driver informed about vehicles that may be approaching to pass, and also identifying cargo securement or load security issues.

Remember that the load driver needs to know where you are. Watch lines of sight or communicate when moving out of sight.

We suggest using "call signs" for communication during the move. They don't have to be fancy or formal, but agreeing ahead of time on how you will address each other may be helpful, especially when other oversize loads are in the area.

Test the frequencies or channels that were specified in your pre-trip meeting and other trip planning, and make plans for what to do if radio communication is temporarily not available.

4.5 Radio Equipment and Frequencies

Radios to be used should either be 40 channel CB (GRS) radios or 128 channel (minimum) VHF radios. If possible, carry a spare radio and extra batteries. Try to avoid chatter, and politely ask other traffic to do the same, if necessary. Carry information about available frequencies for the area and route you will travel, including where international travel applies.

Cell phones are useful as a backup, either via Bluetooth or at roadside.

If the users will be operating on resource roads then they must use VHF radios so that they can communicate with other vehicles on the roads and report their locations. VHF radios are required on resource roads, and for use of LADD channels, and operators need a license from *Innovation, Science, and Economic Development Canada* (ISED). Information about application for that license can be found on their website.

VHF radio operators require an appropriate license to operate in Canada. CB radio operators do not require a license in Canada or the US. However, they must ensure that their radio equipment is legal in Canada (it should have an ISED approval sticker), and they must operate in it a legal manner (no profanity, no transmitting of sensitive material, etc.)

4.5.1 Radio Channels for VHF Radios

For information about available radio channels, please contact the Spectrum Management and Telecommunication branch of ISED. A listing of local offices and contact information may be found at *their website.*

A trucking firm may have its own radio frequencies, licenced to them by ISED. In that case, it would always make sense to use the frequency licenced to the trucking firm instead of the LADD channels, which can be fairly congested and open to everyone, except for communications intended for traffic around the load. Many pilot cars are equipped with and use two VHF radios for this reason.

It is also possible to use another company's assigned frequency only if:

- The frequency is valid in the area they will be using it.
- They have a letter from the company authorizing their use of the frequency.
- They have updated their ISED licence to include the authorized frequency.